Individual Contribution Report - Madeline Bumpus

In this project, we built deep learning models to classify facial images into four pain levels: no pain, mild, moderate, and severe. My main role was implementing and analyzing two transfer learning models, VGG16 and ResNet50, and doing the literature review to help guide our approach. The goal was to see how well pretrained models could improve performance on this type of problem compared to simpler approaches.

For the literature review, I looked into both traditional and more recent approaches to facial pain detection to understand how this problem has been tackled before. Earlier work mainly focused on using Facial Action Coding System (FACS) features, where specific facial muscle movements (like brow lowering or lip tightening) are used to detect pain. More recent research relies heavily on deep learning, especially convolutional neural networks, since they can automatically learn useful features from images without manual feature engineering. I also found that transfer learning is commonly used in this area because pretrained models tend to perform much better on smaller datasets. Some newer approaches also use attention mechanisms or focus on specific regions of the face, like the mouth or eyes, to improve performance. Overall, the literature helped justify our use of pretrained CNNs and motivated exploring models that capture both global and localized facial features. This literature review directly influenced the choice of transfer learning models and the two-phase training strategy used in this project.

The dataset we used consists of RGB facial images grouped into the four pain categories. Even though the classes are fairly balanced, the task itself is still difficult. A big challenge is that the differences between pain levels—especially mild and moderate—are very subtle. On top of that, there are variations in lighting, head position, and facial orientation, and the labels themselves come from emotion-based data, which introduces some noise.

For my two models, I used the preprocessing pipeline, including resizing, normalization using ImageNet statistics, and data augmentation like random flipping, rotation, and color jitter designed by Dhanush. Training was done using a weighted cross entropy loss function. I used accuracy, F1-score, and confusion matrices to evaluate performance, and I compared the models to understand where each one performed well or struggled.

For VGG16, I started with a pretrained model and replaced the final classifier so it could handle four classes instead of the original ImageNet outputs. I added dropout and additional fully connected layers to help reduce overfitting. The training was done in two phases: first, I froze the convolutional layers and trained only the classifier, and then I unfroze the whole network and fine-tuned it with a lower learning rate. This model ended up performing the best overall, with about 79% accuracy and F1 score. It learned quickly at the beginning and then stabilized, although there was some overfitting toward the end. The confusion matrix showed strong

performance overall, with most predictions falling along the diagonal, but there was still some confusion between similar classes like mild and moderate.

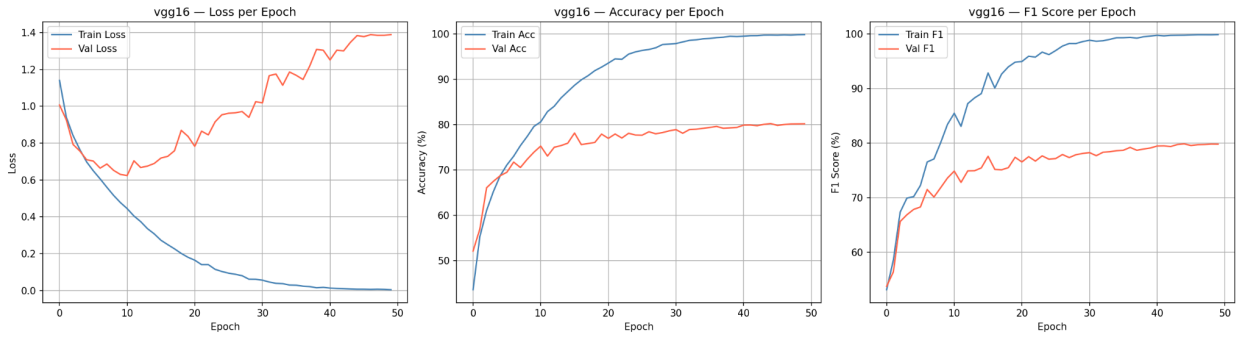

Figure 1: VGG Training Curves

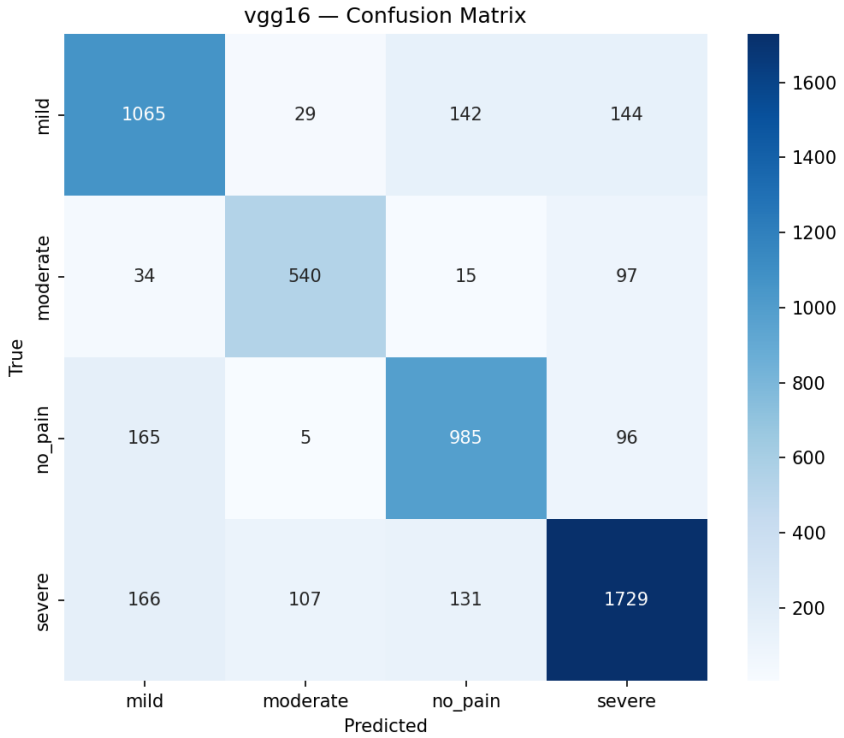

Figure 2: VGG Confusion Matrix

The ResNet50 model followed a similar setup, but with a different architecture that uses residual connections to help train deeper networks. I again replaced the final layer, added dropout, and used the same two-phase training approach. The results were slightly lower than VGG16, with around 77% accuracy and F1 score, but still strong overall. The training process was smooth and stable, and the model showed good generalization, though it also had some confusion between neighboring classes.

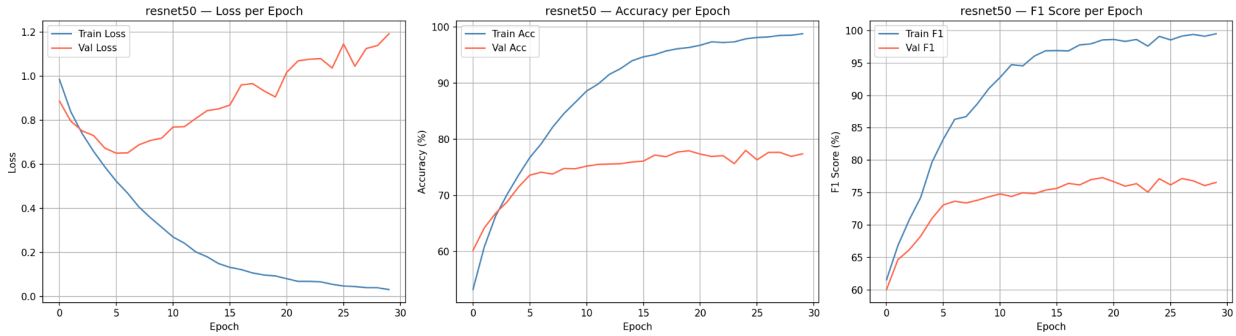

Figure 3: ResNet50 Training Curves

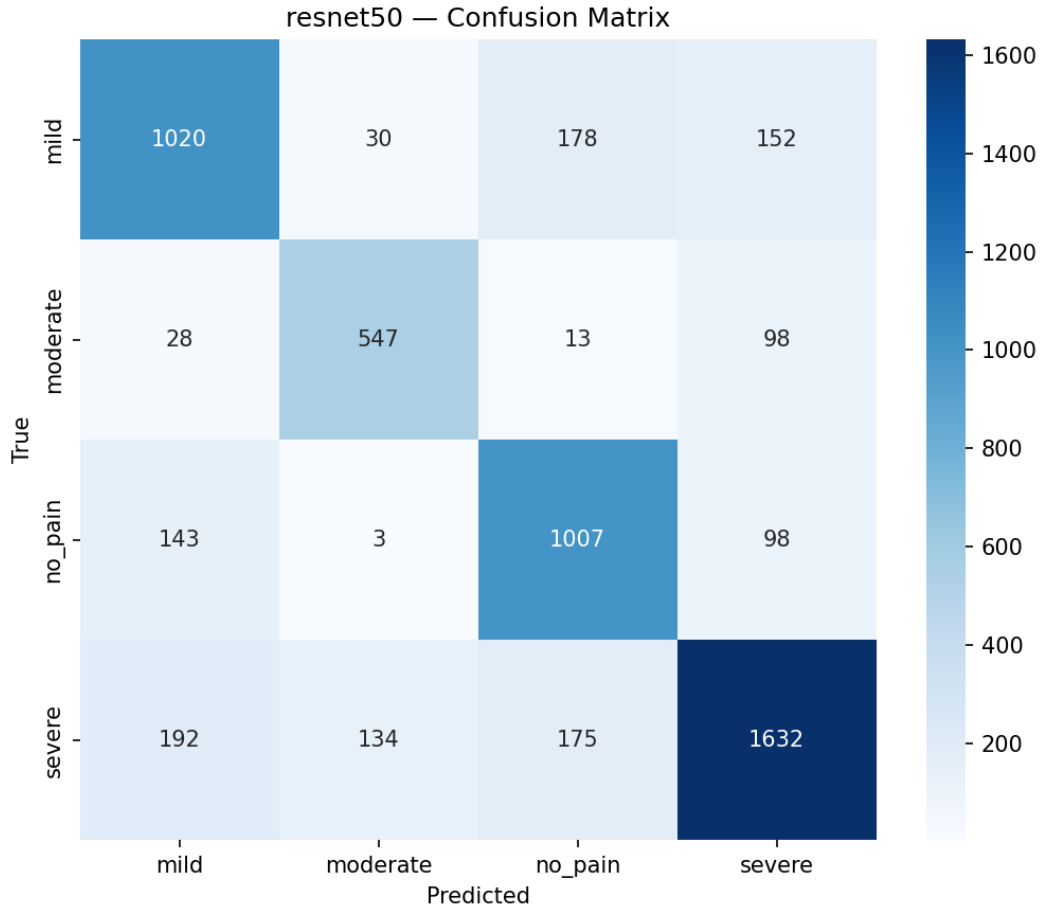

Figure 4: ResNet50 Confusion Matrix

Comparing the two, VGG16 performed slightly better, but both models were a big improvement over the baseline CNN trained from scratch by Simintaj. This really shows how important transfer learning is for this kind of task, since the pretrained models already have strong feature extraction capabilities. The biggest remaining issue across both models was distinguishing between subtle pain levels, which is expected given how similar those facial expressions can be.

Model	Accuracy	F1 Score
VGG16	~79%	~79%
Resnet50	~77%	~77%

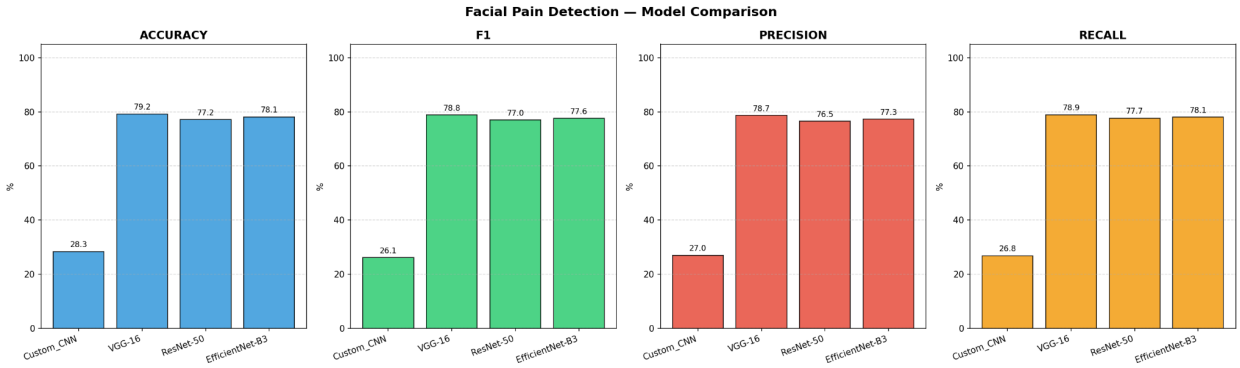

Figure 5: Model Comparison

Some of the main challenges in this project were dealing with subtle differences between classes, working with labels that weren't originally designed for pain detection, and making sure the models generalized well across different image conditions. Even with these challenges, the models were able to learn meaningful patterns, which we confirmed using Grad-CAM visualizations that showed the models focusing on important facial regions like the eyes and mouth.

In terms of code contribution, some parts of the implementation follow standard PyTorch transfer learning patterns, like loading pretrained models, freezing layers, and modifying the final classifier. For VGG16, about 30% of the code comes from common examples, while around 70% was customized by me for this project. For ResNet50, about 25% follows standard patterns, and about 75% was my own work, including model modifications, training setup, preprocessing, and evaluation. Overall, most of the code was adapted specifically for this project rather than copied directly.

Overall, this project showed that transfer learning is very effective for facial pain classification, with VGG16 performing the best in our setup. The two-phase training strategy also played an important role in getting good results. In the future, this could be improved by

using multi-input models (like combining full face and specific regions), working with more accurate clinical datasets, or incorporating video data to capture changes over time.

Bibliography

Brett, K., & Severn, M. (2023). Facial Analysis Technology for Pain Detection: A Potentially Useful Tool for People Living With Dementia: CADTH Horizon Scan. In *PubMed*. Canadian Agency for Drugs and Technologies in Health. https://www.ncbi.nlm.nih.gov/books/NBK595121/

Chen, Z., Ansari, R., & Wilkie, D. (2026). *Welcome To Zscaler Directory Authentication*. Harvard.edu. https://ui.adsabs.harvard.edu/abs/2018arXiv181107988C/abstract

Lucey, P., Cohn, J. F., Lucey, S., Matthews, I., Sridharan, S., & Prkachin, K. M. (2009). Automatically detecting pain using facial actions. *Europe PMC (PubMed Central)*. https://doi.org/10.1109/acii.2009.5349321

Tan, C. W., Du, T., Teo, J. C., Chan, D. X. H., Kong, W. M., & Sng, B. L. (2025). Automated pain detection using facial expression in adult patients with a customized spatial temporal attention long short-term memory (STA-LSTM) network. *Scientific Reports*, *15*(1). https://doi.org/10.1038/s41598-025-97885-5